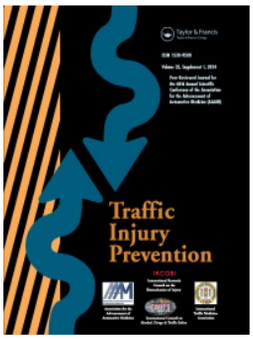

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Comparison of Waymo rider-only crash data to human benchmarks at 7.1 million miles

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ABSTRACT

Objectives: This article examines the safety performance of the Waymo Driver, an SAE level 4 automated driving system (ADS) used in a rider-only (RO) ride-hailing application without a human driver, either in the vehicle or remotely.

Methods: ADS crash data were derived from NHTSA's Standing General Order (SGO) reporting over 7.14 million RO miles through the end of October 2023 in Phoenix, Arizona, San Francisco, California, and Los Angeles, California, and compared to human benchmarks from the literature.

Results: When considering all locations together, the *any injury reported* crashed vehicle rate was 0.6 incidents per million miles (IPMM) for the ADS vs. 2.80 IPMM for the human benchmark, an 80% reduction or a human crash rate that is 5 times higher than the ADS rate. *Police-reported* crashed vehicle rates for all locations together were 2.1 IPMM for the ADS vs. 4.68 IPMM for the human benchmark, a 55% reduction or a human crash rate that was 2.2 times higher than the ADS rate. *Police-reported* crashed vehicle rate reductions for the ADS were statistically significant when compared in San Francisco and Phoenix, as well as combined across all locations and the any-injury-reported reductions were statistically significant in San Francisco and in all locations. The *any property damage or injury* comparison had statistically significant decreases in 3 comparisons but also nonsignificant results in 3 other benchmarks. When excluding ADS crashes with a delta-V less than 1 mph (a measure of sensitivity to lower reporting threshold), about half of the ADS collisions were excluded, resulting in comparisons that showed a large statistically significant reduction in all comparisons except for one comparison from San Francisco.

Conclusions: The statistically significant reductions in *police-reported* and *any injury reported* crash rates indicate a promising positive safety impact of ADS. The direction and significance of comparisons in the *any property damage or injury* outcome group are inconclusive due to difficulties in estimating a matching human benchmark. More research is needed on defining *any property damage or injury* benchmarks with clear lower reporting thresholds to reduce the systematic uncertainty in the benchmark rates. Together, these crash rate results contribute to the continuous growth in confidence, together with other methodologies, in a safety case approach.

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Automated driving systems; safety impact analysis; traffic safety; crash data

Introduction

This article describes crash results from an in-field deployment of a rider-only (RO) automated driving system (ADS) ride-hailing service as part of the safety determination life cycle. The in-field crash results may serve as one important factor in confirming design elements and predictions done in earlier iterations of this ADS safety determination life cycle, as shown in Figure 1 (Favarò, Fraade-Blanar et al. 2023). While a system is being developed and before starting RO operations or when considering an update to an existing RO deployment, only prospective methods that predict expected performance can be applied. In a prospective method—for example, in simulated deployments (see Webb et al. [2020] or Favarò, Victor et al. [2023])—simulation is used to predict an ADS crash rate and compare that rate to a benchmark. In this approach, benchmarks can be set for different severity levels that are determined based on a crash severity model, because outcomes such as injury are not available from simulation. Retrospective analyses based on crash outcomes like those presented in this article can be used to complement and/or confirm these prospective methods that are used in an ADS readiness determination (Webb et al. 2020).

Retrospective examination of crash performance against a human benchmark contributes to the continuous growth in confidence of an ADS safety case. Retrospective studies cannot be used to make safety determinations prior to deployment. Therefore, an ADS with a crash involvement rate

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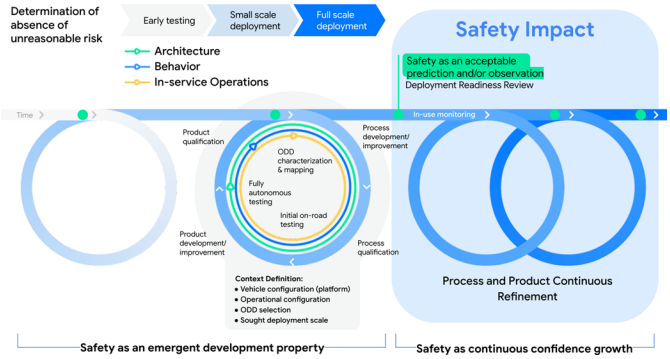

Figure 1. Visualization of Waymo's safety determination life cycle (adapted from figure 6, Favarò, Fraade-Blanar et al. 2023).

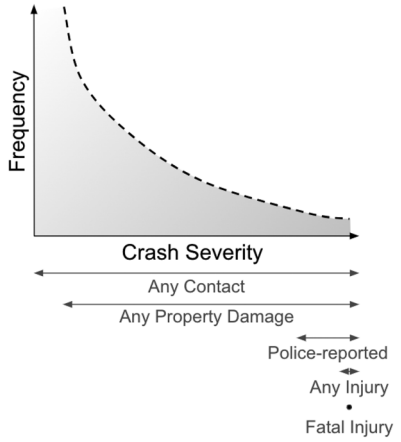

Figure 2. Conceptual relationship between different crash severity reporting thresholds. The arrows below the graph show the hypothetical range of severities included in each outcome group.

lower than the general driving population does not, by itself, allow for a determination that the ADS has an absence of unreasonable risk. Assessment methodologies that use other benchmarks are important in addition to retrospective crash involvement (Webb et al. 2020). The broader safety case approach considers a range of potential issues that can arise during ADS ride-hailing operations, including measures of the ADS's drivership, or the realization of expected driving behaviors that position the ADS as a good citizen of the road. Evaluation of drivership can extend beyond crash risk.

The data sources most commonly used to understand traffic safety historically have been derived from police-reported crash databases, like those maintained by the NHTSA and individual states. Recently, ADS manufacturers have been required to report crashes, albeit with a different reporting standard than that used in police-reported crash databases. Since July 2021, the NHTSA Standing General Order (SGO) 2021-01 has required ADS developers to report all crashes meeting certain criteria, which are then published publicly (NHTSA 2023). The reporting requirements for ADS manufacturers is to report any crashes that the manufacturer knows to have occurred or that are alleged to have occurred and that result in "any property damage, injury, or fatality"; occur on a public road; and in which the ADS was engaged at any time during the 30 s immediately prior to the crash through the conclusion of the crash (NHTSA 2023).

A retrospective safety impact assessment requires calibration of benchmark human and ADS crash and vehicle miles traveled (VMT) data sources to make a valid comparison. Given public accessibility of both human and ADS crash and VMT data, researchers have the opportunity to begin evaluating ADS safety performance. Working with these public crash and VMT data sources, however, has certain challenges. Several reviews of the literature have found that many of the early studies comparing ADS and human benchmarks introduced biases (Young 2021; Scanlon et al. 2024).

One of the most important considerations when matching benchmark and ADS crash and VMT data sources is the inclusion criteria of the data sources being compared (i.e., to correct for surveillance bias). Figure 2 shows the conceptual relationship between different crash severity reporting thresholds relative to their frequency of occurrence. The distribution shows hypothetical frequency by crash severity (e.g., delta-V or impact speed). Below the frequency distribution, there are ranges of severity that correspond to different reporting thresholds for crash data sources. The NHTSA SGO data are of the "any property damage" reporting threshold. Most human crash databases such as NHTSA's Crash Report Sampling System (CRSS) database and state crash databases are derived from crashes reported to police and/or through insurance claims. A subset of police-reported crashes can result in a reported injury (i.e., minor to fatal injuries), and a smaller subset of those *any injury reported* crashes result in fatal injuries. Given these different reporting thresholds, it is not appropriate to compare ADS crash data like the NHTSA SGO, which includes outcomes at the any property damage level, directly to human police-reported crash databases like national or state crash databases.

In addition to reporting thresholds, another form of bias comes from reporting bias due to underreporting. Many human crashes are not reported to police or are reported but no police report is filed. In the United States, 60% of property-damage-only (PDO) and 32% of injury crashes are not reported to police (Blincoe et al. 2023). In contrast, ADS fleet operators are required to report almost every crash, and the large number of sensors combined with operational procedures results in almost all contacts being discovered and reported. Finally, the driving environment and/or vehicle characteristics of a data source can impact the crash rate (selection bias). Factors such as population density (rural vs. urban), roadway type (surface street vs. freeway), and vehicle use type (commercial heavy vehicles, passenger vehicles, ride-hail vehicle) can all affect crash rates.

Due to these potential challenges that introduce bias into comparisons of benchmark and ADS crash data, different approaches are needed to form valid comparisons depending on the nature of the human benchmark and ADS data sources. For example, if a benchmark that is derived from crashes reported to police is compared to ADS data that include any property damage, there is a need to either apply some underreporting adjustment or restrict the ADS data to only include police-reported crashes. For ease of writing, we will use "underreporting" to refer to both crashes that meet the reporting threshold but were not reported and crashes

that do not meet the reporting threshold. See Scanlon et al. (2024) for a more thorough discussion of these topics. An underreporting adjustment to police-reported crashes is the approach Scanlon et al. (2024) took to estimate crash rates for any property damage. Alternatively, another human data source used for generating a benchmark is naturalistic driving study (NDS) databases; in naturalistic driving studies, vehicles are equipped with sensors such as accelerometers and cameras to record driving. Using a combination of algorithmic triggers (e.g., hard braking or swerving) plus manual review to find crashes based on the video data, an observed crash rate can be found, such as those reported in Blanco et al. (2016) and Flannagan et al. (2023).

Generating a valid comparison of human and ADS crash rates that adjusts for the abovementioned biases is of great importance at this point in time. Larger ADS deployments in recent years have led to inquiries from the general public, public officials, and researchers about the safety of current ADS fleets. All stakeholders have an interest in understanding the current state of ADS safety performance through as many objective measures as are available, which include retrospective crash rates. If biases are present in the data that either inflate or deflate the ADS or benchmark crash rates, the conclusions may lead to incorrect public perceptions of ADS safety and questionable policy recommendations or decisions. As pointed out by Scanlon et al. (2024, appendix A.1.2) and Young (2021), past studies have varying degrees of biases in their analyses. In addition, nonacademic studies, either in the popular press or by government agencies, are being done. All data sources, including both human and ADS sources, have limitations. This article aims to present an analysis that uses publicly available ADS and benchmark data to advance the understanding of ADS safety so that stakeholders can make informed decisions. This analysis is also a step toward further harmonization and standardization of retrospective safety impact analysis for ADS fleets.

In the field of traffic safety, the prevention of severe crashes is generally prioritized. One of the paradoxes of the study of crashes, however, is that severe outcomes are less frequent than low-severity outcomes. This is true for several types of outcomes. For example, crashes with low property damage dollar values outnumber those with high dollar values and minor injuries outnumber serious injuries. Given that ADS vehicles drive fewer miles relative to human-driven vehicles, the ADS crash data are overwhelmingly PDO crashes. As discussed above, it is difficult to construct comparable benchmarks for any property damage crashes owing to differences in reporting thresholds and underreporting. Because of the great public interest, this article will attempt to construct a benchmark and compare it to ADS crashes for an *any property damage or injury* outcome despite the challenges. Because the most-studied human crash data are at either the *police-reported* or *any-injury-reported* outcome level (or greater), these data sources are more well understood and are likely to have less systematic biases present compared to *any property damage or injury* benchmarks. The drawback of *police-reported* and *any injury reported* outcome levels is that there will be fewer ADS crashes observed, leading to potentially larger confidence intervals (CIs). Because more serious outcome crashes often have different characteristics than less severe outcome crashes, it is possible that the safety impact and/or the confidence in the impact of an ADS may be different for different outcome levels. See Appendix A.4 (online supplement) for further discussion on considerations for safety impact assessment for serious injury and fatal crashes.

The purpose of this article is to present Waymo ADS crash rates from data reported according to NHTSA's SGO and compare to benchmark rates from comparable human benchmark sources published in the literature. Comparisons were made for crashes involving *any property damage or injury*, *police-reported*, and *any injury reported* outcomes. Because the SGO data are publicly available, the analyses provided herein can be readily reproduced and developed by other researchers.

Methods

Matching of benchmark and ADS data

Due to the data collection differences between human and ADS data, there is no set of data that represent the perfect comparison between human and ADS crash rates. Therefore, this article aimed to develop several comparable sets of human and ADS crash rates using data published from the literature and NHTSA SGO data.

The Waymo RO ride-hailing service in Phoenix, San Francisco, and Los Angeles operates in fixed geographic areas in these cities, operating 24 h a day and 7 days a week. The operational design domain (ODD) includes non–limited access roads with speed limits up to 50 mph and parking lots without restrictions on maneuvers. The ODD does not include severe weather conditions, such as thick fog, heavy rain, or blowing sand, but does include light rain or light fog. The ODD for the Waymo RO service has changed slightly over the years as the service expanded. The ODD features have remained largely consistent since 2022, which also corresponds to the majority of the RO miles accumulated during the study period (through October 2023). Therefore, the Waymo RO crashes described in this section were compared to crashes that occurred in the same geographical areas. The selection of human benchmarks is discussed in detail in the following section.

Because NHTSA SGO ADS crashes are a mandated and publicly available data source, Waymo's NHTSA SGO reportable crashes became the basis for the incidents examined in this study. The SGO reporting is required in all states and thus is the most complete crash reporting required for an ADS fleet operator. Manufacturers do not report miles driven as part of the SGO data. Therefore, this study analyzed only Waymo RO crashes because these miles were the only ones available to the authors. All crashes involving Waymo operating in RO through the end of October 2023 were included. SGO data were downloaded from NHTSA's website in December 2023.

Table 1 lists the ADS data and human benchmarks compared in this study, and Table 2 lists the benchmark crash rates. Comparisons were grouped into 3 outcome groups: *Any property damage or injury*, *police-reported*, and *any injury reported* outcomes. Details of each outcome group are discussed below.

Table 1. Description of comparable ADS and human benchmarks data sources.

Outcome group	Waymo ADS data	Human benchmark source
Any property damage or injury	All NHTSA SGO crash—In-transport vehicles	Scanlon et al. (2024) any property damage or injury, Blincoe-adjusted
	NHTSA SGO crashes—In-transport vehicles excluding low delta-V (<1 mph)	Scanlon et al. (2024) any property damage or injury, Blincoe-adjusted
	All NHTSA SGO crashes in San Francisco	Flannagan et al. (2023)
	All NHTSA SGO crash in all locations	Blanco et al. (2016)
Police-reported	All NHTSA SGO crash—In-transport vehicles with police report	Scanlon et al. (2024) police-reported unadjusted
Any injury reported	All NHTSA SGO crash—In-transport vehicles with reported injury	Scanlon et al. (2024) any injury reported, Blincoe-adjusted
	All NHTSA SGO crash—In-transport vehicles with reported injury	Scanlon et al. (2024) any injury reported, observed

Table 2. Crashed vehicle rates and VMT for human benchmarks.

		Human benchmark IPMM and VMT (millions)			
Outcome group	Human benchmark source	National	Phoenix	San Francisco	Los Angeles
Any property damage or injury	Scanlon et al. (2024), Blincoe-adjusted any property damage or injury	9.40 2,140,140	9.43 24,865	10.49 862	6.84 29,952
	Flannagan et al. (2023) ride-hailing NDS	N/A	N/A	64.9 5.612	N/A
	Blanco et al. (2016) SHRP-2 NDS	20.2 34	N/A	N/A	N/A
Police-reported	Scanlon et al. (2024), observed police-reported	4.10 2,140,140	4.31 24,865	5.86 862	3.39 29,952
Any injury reported	Scanlon et al. (2024), observed any injury reported	1.21 2,140,140	1.24 24,865	3.98 862	1.54 29,952
	Scanlon et al. (2024), Blincoe-adjusted any injury reported	1.76 2,140,140	1.81 24,865	5.82 862	2.26 29,952

This study used 2 types of data sources to match *any property damage or injury* benchmarks to the ADS data. First, police-reported crash data from national (the NHTSA CRSS and Fatality Analysis Reporting System) and state crash databases were adjusted for human underreporting in Scanlon et al. (2024) using underreporting adjustments for PDO and injury crashes from Blincoe et al. (2023). The Blincoe et al. (2023) underreporting adjustments were developed using a combination of national police-reported data, insurance claims data, and a nationally representative phone survey. These national underreporting estimates for PDO and injury crashes were then applied to national and state crash data from 2022 in Scanlon et al. (2024). The state data were restricted to the counties where ADS deployments are currently occurring (San Francisco, California, Maricopa, Arizona, and Los Angeles, California). Both the state and national data were also restricted to crashes involving passenger vehicles and occurring on surface streets to match the operating conditions of the ADS. The human VMT data in Scanlon et al. (2024) were sourced from the Federal Highway Administration for national and state data using the same conditions (passenger vehicles on surface street roads). Second, observed crash rates from NDS were used. Flannagan et al. (2024) derived crash rates from a fleet of vehicles mostly used for ride-hailing on surface streets in San Francisco; these rates were then compared to ADS crashes in San Francisco. Blanco et al. (2016) derived a crash rate using the SHRP-2 NDS, which included personally owned passenger vehicles that were operated on all road types.

For the comparison of Waymo's RO crash rate to available human benchmark crash rates, the following subsets of Waymo's SGO-reported crashes were identified: (1) Those crashes reported by Waymo and without an autonomous specialist behind the wheel (RO crashes), (2) in-transport vehicles (defined as crashes where the Waymo vehicles were not parked in a parking spot), and (3) those where the Waymo vehicle was impacted (defined as those crashes where the Waymo vehicle struck or was struck by another road user or object). In those ADS crashes with an autonomous specialist seated in the driver's seat, it is difficult to disambiguate the contributions of the ADS and the autonomous specialist. Therefore, this analysis excluded those crashes with an autonomous specialist present behind the wheel, because RO crashes are most representative of the intended autonomous ride-hailing service. The crashed vehicle count from the police-reported benchmark data included only in-transport, impacted vehicles. Therefore, the *any property damage or injury* benchmarks from Scanlon et al. (2024) were compared with all SGO-reported ADS crashes, except those where the ADS vehicle was not in transport or the ADS vehicle was not impacted. Because the NDS data sources described in Flannagan et al. (2023) and Blanco et al. (2016) did not specify whether not-in-transport NDS vehicles were excluded, we compared these NDS data to all reported SGO ADS crashes excluding when the ADS vehicle was not impacted. All SGO collisions that met the conditions were included in the study, regardless of crash mode or contacted object.

In addition to examining all NHTSA SGO-reported crashes, results were reported for a data set that excluded minor crashes, which we will refer to as "SGO crash excluding low delta-V." Although the benchmark data were either adjusted for underreporting or used NDS data designed to find most contacts, there is still uncertainty about the lower reporting threshold in the benchmark data and

Table 3. Waymo mileage blended crashed vehicle rates and VMT for human benchmarks from Scanlon et al. (2024).

Outcome group	Human benchmark source	Human benchmark IPMM
Any property damage or injury	Blincoe-adjusted	9.67
Police-reported	Observed police-reported	4.68
Any injury reported	Observed	1.92
	Blincoe-adjusted	2.80

underreporting corrections. This result excluding low delta-V ADS crashes was presented alongside the larger all SGO-reported crashes group to investigate the sensitivity of lower reporting threshold on the comparison to the benchmark. We defined these minor contacts as those in which a crash reconstruction model found that both vehicles had a change in velocity (i.e., a delta-V) of less than 1.0 mph. Crash delta-V was determined by a crash model (for vehicle-to-vehicle crashes) or estimated by collision reconstruction techniques (for fixed object crashes).

The human crash benchmarks reported in Scanlon et al. (2024) for Phoenix, San Francisco, and Los Angeles were combined proportional to the miles driven in the Waymo RO service, as shown in Table 3. This mileage blended average was compared with the ADS crashed vehicle rate aggregated in the 3 locations. For the purposes of CIs, the mileage blended benchmark used 19,002 million VMT, which is the average benchmark VMT proportional to the miles driven in the Waymo RO service.

The second comparison made in this study was between ADS NHTSA SGO crashes with a police report filed and the observed human police-reported rate. Police-reported and alleged injury crashes were identified from the SGO-reported crashes using fields in the NHTSA SGO data. The human police-reported rates in Scanlon et al. (2024) did not have an underreporting correction applied. Third, ADS NHTSA SGO crashes where any injuries were reported were compared to the human injury benchmark from Scanlon et al. (2017), both adjusted for underreporting using Blincoe et al. (2023) and unadjusted. Because *police-reported* and *any injury reported* crashes are relatively rare, most human NDS-based benchmarks have few of these crashes and do not attempt to estimate rates. Therefore, the only comparable benchmark data sources available were the police-reported and any injury reported rate estimates presented in Scanlon et al. (2024).

This study did not compare the Waymo ADS performance to all human benchmarks developed in Scanlon et al. (2024). Scanlon et al. (2024) also presented an *any property damage or injury* crash rate using a PDO underreporting estimated from the SHRP-2 NDS developed in Blanco et al. (2016). This current study did not compare the ADS to this Blanco-adjusted benchmark because Blanco et al. (2016) only attempted to estimate unreported crashes (those that met the reporting threshold but were not reported) where the Blincoe et al. (2023) underreporting amount accounted for both unreported crashes and crashes that did not meet the reporting threshold (Scanlon et al. 2024). Scanlon et al. (2024) also presented a tow-away benchmark (i.e., any vehicle was towed from the scene due to damage). This study did not compare the ADS to this tow-away benchmark because ADS vehicles are likely towed from the scene more often than human-driven vehicles for reasons that are unrelated to the amount of damage caused by the crash (e.g., due to damage to sensor covers). Finally, the ADS was not compared to the airbag deployed, suspected serious injury+ (incapacitating or fatal injuries), or fatal injury benchmarks from Scanlon et al. (2024) because there are not yet sufficient miles in the ADS data to draw conclusions about these more rare outcomes (see statistical power analysis of Kalra and Paddock [2016] and Scanlon et al. [2024], discussed further in Appendix A.4 of the current article).

Details of the ADS data selection criteria, the collision model, and a list of all crash incidents included for analysis in this article are included in Appendix A.1 (see online supplement).

Confidence intervals

Two types of CIs are presented in this article. First, CIs are estimated for incidents per million miles (IPMM) using a Poisson exact model. The CI, IPMM, can be computed as

$$IPM \in \left[qgamma(\alpha/2, n, 1)/m, qgamma(1-\alpha/2, n+1, 1)/m \right] \quad (1)$$

where *qgamma* is the quantile function of the gamma distribution (i.e., inverse cumulative distribution function), α is the significance level (e.g., .05), n is the event count, and m is the number of miles driven. Second, CIs for the ratio between the ADS crash rate of Y events over t miles and the benchmark crash rate of X events over s miles were calculated using the method described by Nelson (1970) as

$$\rho \in \begin{bmatrix} s/t * qbetaprime(\alpha/2, Y, X+1), \\ s/t * qbetaprime(1-\alpha/2, Y+1, X) \end{bmatrix} \quad (2)$$

where *qbetaprime* is the quantile function of the beta prime distribution. This formulation of the CIs can be computed in the event of zero event counts in the ADS data (i.e., $Y=0$) but as a result may be more conservative (i.e., wider CIs) in the case of nonzero ADS events counts compared to the Poisson exact model. Appendix A.3 (see online supplement) provides example code and test cases.

Another interpretation of the ratio between ADS rate ($\mu = Y/t$) and benchmark rate ($\lambda = X/s$) is a relative difference in the rate δ,

$$\delta = (\mu - \lambda)/\lambda = \mu/\lambda - 1. \quad (3)$$

Because this percentage change in the rate is a direct product of the ratio, the same CIs described in Eq. (2) can be used to generate CIs for the relative difference.

The national crash rates from Scanlon et al. (2024) were derived from crash counts from the CRSS database, which features a complex survey design for sampling crashes. Standard errors for counts that use the survey design variables are larger than standard errors calculated using conventional methods using just the weighted counts (F. Zhang and Diaz 2020). To investigate the effect of the CRSS

Table 4. Number and incidents per million miles (IPMM) of crashed vehicles for Waymo NHTSA SGO-reported crashes with 95% CIs in Phoenix and San Francisco.

	Phoenix			San Francisco		
Measure	*n*	IPM	CI	*n*	IPM	CI
RO miles (millions)	5.34	—	—	1.76	—	—
SGO-reported	38	7.1	(5.0, 9.8)	34	19.4	(13.4, 27.1)
SGO-reported in-transport	33	6.2	(4.3, 8.7)	29	16.5	(11.1, 23.7)
SGO-reported exclude low delta-V	17	3.2	(1.9, 5.1)	14	8.0	(4.4, 13.4)
SGO police-reported	12	2.2	(1.2, 3.9)	3	1.7	(0.4, 5.0)
SGO any injury reported	3	0.6	(0.1, 3.2)	1	0.6	(<0.1, 3.2)

Table 5. Comparison of Waymo ADS and human benchmark crashed vehicle rate for crashes with any property damage or injury, all NHTSA SGO-reported crashes.

Human benchmark	Location	Human IPMM	ADS IPMM	ADS-to-human rate ratio	95% Confidence intervals	
Scanlon et al. (2024), Blincoe-adjusted	Phoenix	9.43	6.2	0.65*	0.43	0.96
	San Francisco	10.5	16.5	1.57	0.99	2.37
	Total—Mile blend	9.67	8.8	0.91	0.67	1.20
	Total—National average	8.94	8.8	0.99	0.72	1.30
Flannagan et al. (2023), ride-hailing NDS	San Francisco	64.9	19.4	0.30*	0.19	0.44
Blanco et al. (2016), SHRP-2 NDS	All locations	20.2	10.2	0.51*	0.38	0.67

*Statistically significant at 95% confidence.

Table 6. Comparison of Waymo ADS and human benchmark crashed vehicle rate for crashes with any property damage or injury, SGO-reported crashes excluding low delta-V.

Human benchmark	Location	Human IPMM	ADS IPMM	ADS-to-human rate ratio	95% Confidence intervals	
Scanlon et al. (2024), Blincoe-adjusted	Phoenix	9.43	3.2	0.34*	0.18	0.57
	San Francisco	10.5	8.0	0.76	0.38	1.36
	Total—Mile blend	9.67	4.5	0.46*	0.30	0.68
	Total—National average	8.94	4.5	0.50*	0.32	0.74
Flannagan et al. (2023), ride-hailing NDS	San Francisco	64.9	8.0	0.12*	0.06	0.22
Blanco et al. (2016), SHRP-2 NDS	All locations	20.2	4.5	0.22*	0.14	0.33

*Statistically significant at 95% confidence.

variance estimates on the CIs for the rate ratio, the authors constructed CIs using a parametric bootstrap using the standard error for the benchmark crash counts estimated using the survey design variables. The rate ratio CIs computed using the parametric bootstrap method were narrower than those computed using the method described in Eq. (2). This result suggests that the method from Nelson (1970) is more sensitive to the other parameters in the rate ratio calculation (i.e., the count of ADS events) than a parametric bootstrap method. The Nelson (1970) method is more conservative (i.e., produces larger CIs) relative to the parametric bootstrap method. For this reason, the CIs described in Eq. (2) were used to construct rate ratio CIs for the national benchmark comparisons.

Results

In total, this study considered the first 7.14 million miles of driving by the Waymo RO service, which drove 5.34 million miles in Phoenix, 1.76 million miles in San Francisco, and 0.0467 million (i.e., 46,700) miles in Los Angeles. Table 4 shows the number, IPMM, and 95% CIs for the different event measures examined in this study. Results for Los Angeles, which are based on a single collision in the *any property damage or injury* category, are listed in Appendix A.2 (see online supplement).

Tables 5 and 6 report the rate ratio of ADS to human crashed vehicle rates and 95% CIs for the *any property damage or injury* outcome group. Asterisks indicate statistical significance (i.e., the 95% CIs do not contain a 1.0 rate ratio). The human benchmarks are compared to 2 subsets of the ADS crash data: All NHTSA SGO-reported crashes (Table 5) and SGO-reported crashes excluding low delta-V (<1 mph; Table 6). The former is the broadest definition of *any property damage* without any lower threshold for property damage. The latter is shown to demonstrate the sensitivity of the results to the minimum inclusion criteria on the ADS side, because the human benchmarks all had different inclusion criteria for *any property damage or injury*. Of the 63 SGO-reported transport crashes, 32 crashes (or 51%) had a delta-V greater than 1 mph.

Table 7 shows the ADS-to-human rate ratio and 95% CIs for *police-reported* and *any injury reported* crashed vehicle rates. The ADS had a lower *police-reported* and Blincoe-adjusted *any injury reported* crashed vehicle rate. The comparison was statistically in all locations for the police-reported outcome and in San Fransico and when considered in aggregate for the any-injury-reported outcome. The observed *any injury reported* comparison was not statistically significant in Phoenix and when compared to a national average but significant in San Francisco and in the mileage blended comparison. The comparison of the ADS to the observed *any injury reported* benchmark serves as an analysis of the sensitivity of the results to the Blincoe injury underreporting estimate. The results show that even when not accounting for known underreporting in human data, the point

Table 7. Comparison of ADS and human benchmark crashed vehicle rate for police-reported and any injury crashes.

ADS events	Human benchmark	Location	Human IPMM	ADS IPMM	ADS-to-human rate ratio	95% Confidence intervals	
Police-reported	Scanlon et al. (2024), observed police-reported	Phoenix	4.31	2.2	0.52*	0.24	0.97
		San Francisco	5.86	1.7	0.29*	0.05	0.95
		Total mileage blended	4.68	2.1	0.45*	0.23	0.78
		Total national average	4.1	2.1	0.51*	0.26	0.90
Any injury reported	Scanlon et al. (2024), observed any injury reported	Phoenix	1.24	0.6	0.45	0.45	1.47
		San Francisco	3.98	0.6	0.14*	0.002	0.91
		Total mileage blended	1.91	0.6	0.29*	0.06	0.82
		Total national average	1.21	0.6	0.46	0.10	1.30
	Scanlon et al. (2024), Blincoe-adjusted any injury reported	Phoenix	1.81	0.6	0.31	0.05	1.01
		San Francisco	5.82	0.6	0.10*	0.001	0.62
		Total mileage blended	2.80	0.6	0.20*	0.04	0.56
		Total national average	1.76	0.6	0.32*	0.07	0.90

*Statistically significant at 95% confidence.

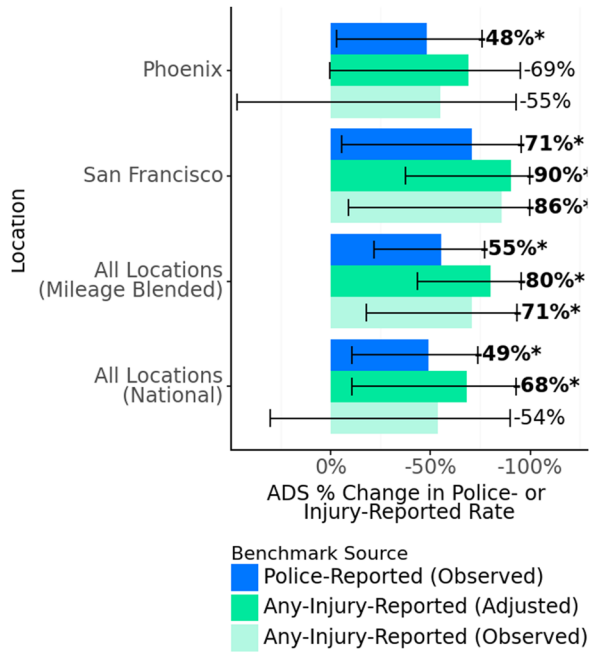

Figure 3. Percentage reduction for the Waymo ADS in *police-reported* and *any injury reported* crashes compared to a human benchmark from Scanlon et al. (2024) in Phoenix, San Francisco, and all locations. Note that Los Angeles data are not plotted because of large error bars. See the results in the online supplement.

estimates still show that the ADS had a lower *any injury reported* crash rate, although not all comparisons were statistically significant due to the low event counts and mileage. Inverting the rate ratios in Table 7, humans have between 1.9 and 3.4 times the rate of *police-reported* crashes and between 3.1 and 10 times the rate of Blincoe-adjusted *any injury reported* crashes as the Waymo ADS. When considering all locations together, the *any injury reported* crashed vehicle rate was 0.6 IPMM for the ADS vs. 2.80 IPMM for the human benchmark, an 80% reduction or a human crash rate that is 5 times higher than the ADS rate. *Police-reported* crashed vehicle rates for all locations together were 2.1 IPMM for the ADS vs. 4.68 IPMM for the human benchmark, a 55% reduction or a human crash rate that was 2.2 times higher than the ADS rate. Figure 3 shows the percentage reduction of the ADS for *police-reported* and *any injury reported* crashed vehicle rates and 95% CIs. This percentage reduction is another common form of the rate ratios shown in Table 7.

Discussion

Comparison to previous studies

This study is one of the first to compare RO ADS crash rates to comparable benchmarks in the geographic locations in which ADS currently operate. In addition to comparing *any property damage or injury* outcomes from SGO-reported crashes, this article compares *police-reported* and *any injury reported* subsets of the SGO data to human benchmarks.

Many previous studies compared early ADS testing data from California, where a human behind the wheel supervised the ADS, sometimes referred to as testing operations (TO) (Schoettle and Sivak 2015; Blanco et al. 2016; Dixit et al. 2016; Favarò et al. 2017; Teoh and Kidd 2017; Banerjee et al. 2018; Goodall 2021). All of these early studies use data prior to 2020 when RO operations started and thus the results from the current study may be more relevant in understanding the current state of the art of RO deployments.

Several studies have examined either more recent TO data or RO data. The human benchmark derived by Flannagan et al. (2023) from ride-hailing NDS was compared to the first 1 million miles of RO driving by the company Cruise, which also operated an RO ride-hailing in San Francisco (L. Zhang 2023). L. Zhang (2023) found a 65% reduction in the *any property damage or injury* outcome category, compared to a 70% reduction found in the current study (0.30 rate ratio in San Francisco compared to the Flannagan et al.'s [2023] benchmark). Chen and Shladover (2024) compared both testing operations and RO crash rates in San Francisco to a human ride-hailing driver crash rate collected as part of data reporting required in California by the California Public Utility Commission (CPUC). The reporting requirement for this CPUC human ride-hailing crash data is not clearly specified, but presumably the requirement is close to the *any property damage or injury* threshold. Chen and Shladover (2024) found that the human ride-hailing crash rate was 15.5 IPMM in 2020 compared to an 14.1 IPMM for just under 1 million miles of Waymo RO vehicles from September 2022 to August 2023, which represents the ADS reducing the crash rate by 9% compared to

the benchmark. No statistical tests were performed by Chen and Shladover (2024) due to the low mileage in the ADS data. The human ride-hailing benchmark in San Francisco reported by Chen and Shladover (2024) is higher than the Scanlon et al. (2024) Blincoe-adjusted *any property damage or injury* benchmark and lower than the Flannagan et al. (2023) ride-hailing NDS benchmark. Due to an unclear lower reporting threshold in the CPUC data used to derive the human ride-hailing benchmark in Chen and Shladover (2024), it is difficult to interpret the discrepancies between the Chen and Shladover (2024) and Flannagan et al. (2023) benchmarks because of the unclear reporting mechanisms in the CPUC data and their implications on underreporting in the human benchmark.

Cummings (2024) compared SGO-reported collisions in California from 2021 to 2022 to a police-reported crash rate. The data used in Cummings (2024) mixed RO and TO data from California. Cummings (2024) compared a Waymo ADS *any property damage or injury* crash (all SGO-reported crashes) to a *police-reported* crash rate derived from national crash and VMT data. As discussed previously, the SGO reporting threshold includes any property damage with no lower reporting threshold, whereas police-reported data require a higher amount of property damage or an injury. Therefore, comparing the ADS SGO crash rate to a police-reported crash rate introduces selection bias that deflates the human benchmark relative to the ADS data. Furthermore, the effects of underreporting in the human data were not accounted for in Cummings (2024), introducing reporting bias that also deflates the human crash rate relative to the ADS crash rate. Due to these biases, Cummings (2024) incorrectly concluded that the Waymo ADS had a statistically significant 4 times higher crash rate than the human benchmark. Indeed, when comparing the Waymo RO *any property damage or injury* rate in San Francisco from Cummings (2024; 16.5 IPMM) to the national observed *police-reported* benchmark (Scanlon et al. [2024]; 4.10 IPMM), the ADS mistakenly appears to have a 4.0 times higher crash rate. The *any property damage or injury* and *police-reported* comparisons done in the current article control for these selection and reporting biases by using benchmarks that attempt to match the reporting threshold and account for underreporting and additionally use benchmarks restricted to passenger vehicles and from the areas the ADS operates in (the *police-reported* crash rate in San Francisco of 5.86 IPMM is 40% larger than the national average of 4.10 IPMM).

The benchmarks and ADS data presented in this article are overall crash involvement rates that do not consider crash contribution of those involved. Di Lillo et al. (2024) compared third-party liability property damage and bodily injury insurance claims rates between the Waymo ADS operations and a human benchmark. A liability claim is a request for compensation when someone is responsible for damage to property or injury to another person, typically following a collision. In Di Lillo et al. (2024), only third-party liability claims that were resolved or were likely to be resolved with a payment were included because these types of claims are a proxy for responsibility. Di Lillo et al. (2024) found that over 3.8 million miles of RO driving, the Waymo RO service had a statistically significant lower property damage and bodily injury claims rate compared to a human benchmark based on aggregate human insurance claims data restricted to ZIP codes where the Waymo service operates. Examining third-party liability claims allows an analysis of the degree to which the ADS contributes to crash causation; that is, the ADS safety impact on other road users. The results of the current study, in contrast, examine the overall crashed vehicle rate regardless of the contribution to the crash's causation. Many of the crashes in the RO data set examined in this article featured the Waymo ADS parked or stopped appropriately at a traffic control device where the ADS's behavior had little or no contribution to the crash. Examining ADS performance from both a crash causation and overall crash rate is important because it provides assessment of (a) performance in crashes where there was some contribution (third-party liability claims), (b) performance in crashes where there was no contribution, and (c) the combination (overall crash rates). Di Lillo et al. (2024) reported that Waymo RO operations reduced property damage liability claims by 76% and bodily injury liability claims by 100%. This magnitude of claims reduction is similar to the magnitude in *police-reported* (49% to 71% reduction) and *any injury reported* (69% and 90% reduction) reductions reported in the current study, which did not consider contribution to causing the crash.

Interpretation of results

Any property damage or injury

The available benchmark crashed vehicle rates for *any property damage or injury* crashes vary widely depending on the method of estimation (i.e., underreporting adjustment of crash data vs. observations in NDS) and the driving environment (e.g., San Francisco ride-hailing NDS vs. national average NDS). Both NDS benchmarks used a 4-level severity scale from the SHRP-2 NDS (Hankey et al. 2016) and included level 1 through level 3 severity crashes in their benchmark (Blanco et al. 2016; Flannagan et al. 2023). In particular, the most frequent level 3 events include events like road departures and curb strikes with an "increased risk element" without vehicle damage, as well as crashes resulting in minor damage. This level 3 definition does not match the SGO reporting requirement of property damage, nor does it likely match the various crash definitions used to construct the underreporting adjustment in Blincoe et al. (2023). This may partially explain why the NDS crash rates are higher than the Blincoe-adjusted rates. Nevertheless, given the available data, it is unclear what magnitude of effect on the *any property damage or injury* benchmark is due to unclear or different reporting thresholds, underreporting differences by location, or the different characteristics of the driving populations (all drivers vs. ride-hail drivers using rental vehicles). For example, Scanlon et al. (2024) found that state data from California had a much lower proportion of reported PDO crashes relative to injury crashes compared with Arizona and national data, which strongly suggests that the national results from Blincoe et al. (2023) may not

generalize to individual states. Blanco et al. (2016) attempted to estimate the rate of collisions that met a reporting threshold but were not reported to police, which resulted in a 84% PDO underreporting amount compared to a 60% underreporting found by Blincoe et al. (2023). Although the Blanco et al. (2016) study had fewer participants and other limitations, the discrepancy suggests there is potential variability in the property damage underreporting amount.

A conservative interpretation of the results of the *any property damage or injury* outcome group is that conclusions cannot be drawn for this outcome group until benchmarks are further developed. The comparisons in the current study do show, however, the possibility of the ADS to reduce crash rates. The ADS reduced crash rates in the comparison of all SGO crashes to a Blincoe-adjusted police-reported rate in Phoenix at a statistically significant level but a higher (but not statistically significant) crash rate for the Blincoe-adjusted rate in San Francisco. The lower PDO reporting in California discussed above suggests that the true crash rate in San Francisco may be higher than that estimated from the police-reported crash data, but to what degree could not be determined. The comparisons of all SGO collisions to the NDS benchmarks from Flannagan et al. (2023) and Blanco et al. (2016) showed a large, statistically significant reduction in crash rate for the ADS. As discussed above, however, there are likely events that did not result in property damage in the NDS data samples (i.e., the level 3 events from the SHRP-2 event definitions), and thus the NDS crash rates are likely inflated to some degree relative to the ADS crash rate. The comparison of SGO collisions excluding crashes with delta-V less than 1 mph showed that many of the ADS collisions were of low severity. As a result, this comparison showed that the ADS had a reduction in crash rate at a statistically significant level in all locations except for San Francisco (which, again, could be partially explained by the property damage reporting discrepancies in California).

To address these discrepancies and difficulties for the *any property damage or injury* benchmarks, there is a need for future research that can more accurately define an objective lower reporting threshold in human crash data than exists today. Such benchmarks would also require standardized methods for reporting the severity of ADS crashes, like the delta-V threshold used in this study. The availability of sensor data enables accurate reconstructions of ADS crashes to estimate their severity (e.g., delta-V or impact speed). Estimating crash severity is often not possible from most human crash databases without detailed reconstructions. There are challenges in generating benchmarks for a specific driving environment using sampled or nonrepresentative databases that do have such crash reconstructions. Another potential source of severity benchmark is to estimate severity from NDS data. Advances in computer vision may enable crash reconstructions from limited vehicle sensor and video data (see, for example, Campolettano et al. 2023).

Police-reported and any injury reported

Observed *police-reported* crashes, in contrast to *any property damage or injury* crashes, may provide a comparison with less inherent systematic variability. Although reporting thresholds and reporting practices may differ by jurisdiction (Scanlon et al. 2024), if observed police reports in the ADS and human benchmark are from the same jurisdictions and time periods, the reporting practices should be comparable in both populations.

This comparison of *police-reported* crashed vehicle rates is likely a conservative comparison for several reasons. Private citizens often do not report police-reportable events (M. Davis & Co 2015). In California, only injury crashes are required to be reported to the state databases, which means some jurisdictions in California may have different PDO police-reported crash reporting practices. Conversely for ADS crashes, this study conservatively used the "Law Enforcement Investigating?" field of the NHTSA SGO reports because the authors have knowledge of Waymo's operational policies, from which it is likely that police took information that could be used to file a report or indicated they would file a report. This field, as Waymo reports to the SGO, is only reported "yes" when law enforcement indicate that they will file a report not only when law enforcement responds to the scene of a crash. Although the *police-reported* outcome is often associated with a certain level of property damage, factors other than property damage may influence whether an ADS fleet operator reports crashes to the police. Future research should investigate police reporting differences between businesses (like an ADS fleet operator) and the general population.

The *any injury reported* outcome group may also serve as a useful comparison between ADS and a benchmark. It has been shown that there is less underreporting in injury crashes compared to crashes with no injuries (Blincoe et al. 2023). Because the magnitudes of underreporting are smaller, the effect of the underreporting estimate on the comparisons may also be lower, as shown by the results of this study. The injury rates, however, can also suffer from similar surveillance biases as the property damage and police report data related to underreporting. The minimum threshold of an injury is often unclear and is dependent on self-reported injury data. Most injuries in the ADS data set were minor and did not require medical treatment. As more RO miles are driven and rarer injury crashes that require medical treatment occur, there is an opportunity to use coded injury scales like the Abbreviated Injury Scale (Association for the Advancement of Automotive Medicine 2016) to more precisely define injury thresholds.

This study compared the crashed vehicle rate of the Waymo RO service to human benchmarks for *any property damage or injury*, *police-reported*, and *any injury reported* outcomes. The results show that the Waymo RO service had a lower *police-reported* and *any injury reported* crashed vehicle rate compared to the benchmarks. When considering all locations together, the *any injury reported* crashed vehicle rate was 0.41 IPMM for the ADS vs. 2.80 IPMM for the human benchmark, an 85% reduction or a human crash rate that is 6.7 times higher than the ADS rate. *Police-reported* crashed vehicle rates for all locations together were 2.1 IPMM for the ADS vs. 4.68 IPMM for the human benchmark, a 55% reduction or a human crash rate that was 2.2

times higher than the ADS rate. Because the *police-reported* and *any injury reported* crashed ADS vehicle rates were much lower than the benchmark rates, the ADS *any injury reported* and *police-reported* crashed vehicle rates were statistically significant over a total of 7.14 million RO miles when compared individually in Phoenix and San Francisco and when compared overall. Although these results are promising, they are based on a small number of ADS events (15 police-reported and 3 any injury reported). Thus, these comparisons may be subject to statistical noise as more miles and events are observed in the future. The comparison for Los Angeles was not statistically significant for zero observed ADS events but few miles. The results for the *police-reported* and *any injury reported* outcome groups, however, show a positive safety impact of the ADS in these types of crashes.

The results are more difficult to interpret for the *any property damage or injury* outcome comparison. The direction of the result and the statistical significance of the comparison depended on which benchmark was used (underreporting adjusted police report or NDS data) and whether low delta-V (<1 mph) crashes were excluded from the ADS comparison. The results of the *any property damage or injury* comparison showed a statistically significant decrease in crashes in Phoenix when the ADS was compared to a police-reported benchmark adjusted for underreporting and in San Francisco and overall when compared to NDS benchmarks. There was an increase in the ADS crash rate when compared to an underreporting adjusted benchmark in San Francisco that was not statistically significant. Half of all ADS crashes had a delta-V less than 1 mph. When excluding ADS crashes with a delta-V less than 1 mph, the decrease in *any property damage or injury* crash rate for the ADS was statistically significant in all comparisons except in the comparison to the underreporting adjusted benchmark from San Francisco. These inconclusive results are largely due to an unclear lower reporting threshold in the human benchmarks. More research is needed to develop benchmarks that have an objective lower reporting threshold to draw conclusions about the *any property damage or injury* outcome group.

This is one of the first studies to compare overall crashed vehicle rates of ADS RO data only (as opposed to a mix of ADS testing with a human behind the wheel) to human benchmarks that also corrected for biases reported in the literature. The comparison of crash rates to benchmarks is part of the safety determination life cycle, which itself is a part of a broader ADS safety case. The results show that the Waymo ADS operating in RO configuration has a lower crashed vehicle rate than human drivers when considering *police-reported* and *any injury reported* crashes. This result provides directional confirmation of the predictions and analyses used prior to deploying RO operations.

Limitations

In addition to the limitations discussed above, this study has several additional limitations. The results in this study examine crashed vehicle rates aggregated by operating location and do not attempt to evaluate crashed vehicle rates within subgroups; for example, by crash partner or crash type. Past studies that reported ADS crash data have found that ADS vehicles may have different proportions of different crash types than the general population (Teoh and Kidd 2017; Victor et al. 2023). A lower aggregate crash rate does not preclude an ADS from having an elevated crash rate in certain types of crashes. As more RO data are collected and statistical power allows, future research should investigate crash rates of more granular crash types.

Although the benchmarks used in this study could adjust for some factors that are related to crashed vehicle rate (e.g., geographical area, types of roads, vehicle type), other pertinent factors were not controlled for. These include time of day adjustments and adjustments for the driving density within the geographic areas. Because the Waymo ADS is being used as a ride-hailing service, the routes the ADS drives are likely differently dispersed than those used by the general driving population and may be biased toward more densely populated areas. There are challenges when matching benchmark crash and VMT data sources that can be adjusted by these parameters, particularly in VMT data. An aggregate-level analysis comparing the ADS to the current crash status quo should include the ADS route choices, even if different from those of comparable human drivers, if the research question (like the one posed in this article) is to determine the safety impact of replacing human driving with ADS vehicle driving.

An ADS vehicle operating in a ride-hailing context is not always occupied by a human (e.g., before or in between picking up passengers). Therefore, when examining injury outcomes, if the ADS vehicle is unoccupied, there would be a reduced risk of injury for a similar human-only crash. As in the route choice discussion above, the objective of this study was to compare the ADS crash involvement to current human drivers, so it is appropriate to include miles where the ADS vehicle is not occupied. More broadly, risk of injury in a motor vehicle crash can be affected by many factors, such as seat belt use and vehicle model year (as a proxy for available safety systems). A risk-based approach that uses an injury risk function to compute probability of an injury outcome regardless of whether an occupant is in the vehicle given some set of inputs can be advantageous in normalizing risk in both the benchmark and ADS crashes for comparison, as has been traditionally done in prospective safety benefit analyses (for example, Kusano and Gabler [2012] and Scanlon et al. [2017]).

The Waymo RO service studied in this article was operating over multiple years on different vehicle platforms and with different software versions. Just as the human driving population has changing characteristics over time, so do certain aspects of the Waymo RO ride-hailing service. The operating environment and territory have expanded during this time period. Starting in the area around Chandler, Arizona, in 2020, today the Waymo RO ride-hailing service operates in an area that includes multiple cities in metropolitan Phoenix and over almost the entire area of San Francisco. A limitation of this study is that the entire history of the Waymo RO crash record was compared to human crash data over different time periods. As ADS fleets continue to increase the number of VMT, additional analyses that are restricted to certain periods of time or platforms could be possible. Many past retrospective studies of safety technologies, however,

group different implementations of the same technology over long time periods to investigate the overall effect of a technology. For example, vehicles from different manufacturers were compared to investigate the effectiveness of first- and second-generation airbags (Olson et al. 2006) and electronic stability control (Riexinger et al. 2019). Thus, aggregate performance of RO ADS is still informative at this point in time.

This study compared the crashed vehicle rate of the Waymo RO service to human benchmarks for *any property damage or injury, police-reported*, and *any-injury-reported* outcomes. The results show that the Waymo RO service had a lower *police-reported* and *any-injury-reported* crashed vehicle rate compared to the benchmarks. When considering all locations together, the *any-injury-reported* crashed vehicle rate was 0.6 incidents per million miles (IPMM) for the ADS vs 2.8 IPMM for the human benchmark, an 80% reduction or a human crash rate that is 5 times higher than the ADS rate. *Police-reported* crashed vehicle rates for all locations together were 2.1 IPMM for the ADS vs. 4.68 IPMM for the human benchmark, a 55% reduction or a human crash rate that was 2.2 times higher than the ADS rate. The ADS *any-injury-reported* and *police-reported* crashed vehicle rates lower than the benchmark rates at a statistically significant level over a total of 7.14 million RO miles when compared individually in Phoenix and San Francisco and when compared overall (except for the *any-injury-reported* comparison in Phoenix). Although these results are promising, they are based on a small number of ADS events (15 police-reported and 4 any injury-reported). Thus, these comparisons may be subject to statistical noise as more miles and events are observed in the future. The comparison for Los Angeles was not statistically significant for zero observed ADS events but few miles. The results for the *police-reported* and *any-injury-reported* outcome groups, however, show a positive safety impact of the ADS in these types of crashes.

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